

Measurement of Capacitance by Schering Bridge

AIM

To Determine the Capacitance of an unknown Capacitor.

APPARATUS REQUIRED:

- Virtual Lab

THEORY:

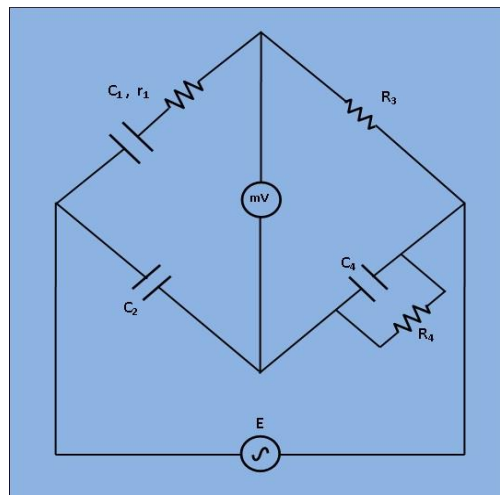


Fig 1: Circuit diagram for measurement of Capacitance by Schering Bridge

Let,

C_1 =capacitor whose capacitance is to be measured.

r_1 = a series resistance representing the loss in the capacitor C_1 .

C_2 = a standard capacitor.

R_3 = a non inductive resistance.

C_4 = a variable capacitor.

R_4 = a variable non inductive resistance.

At balance,

$$(r_1 + \frac{1}{j\omega C_1}) * (\frac{R_4}{j\omega C_4 R_4 + 1}) = \frac{R_3}{j\omega C_2} \dots\dots (1)$$

$$r_1 R_4 - \frac{jR_4}{\omega C_1} = -\frac{jR_3}{\omega C_2} + \frac{R_3 R_4 C_4}{C_2} \dots\dots (2)$$

Or Equating the real and imaginary terms in equa. (2), we obtain

$$r_1 = R_3 * \frac{C_4}{C_2} \dots\dots (3)$$

$$C_1 = R_4 * \frac{C_2}{R_3} \dots\dots (4)$$

And, Two independent balance equations (3) and (4) are obtained if C_4 and R_4 are chosen as the variable elements.

Dissipation factor

$$D_1 = \omega C_1 r_1 \dots\dots (5)$$

PROCEDURE:

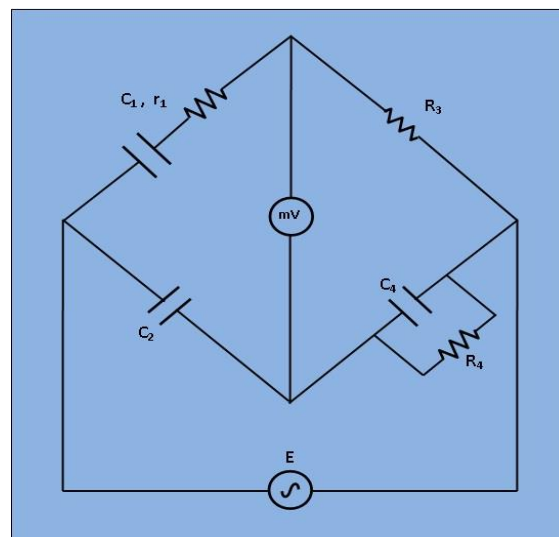
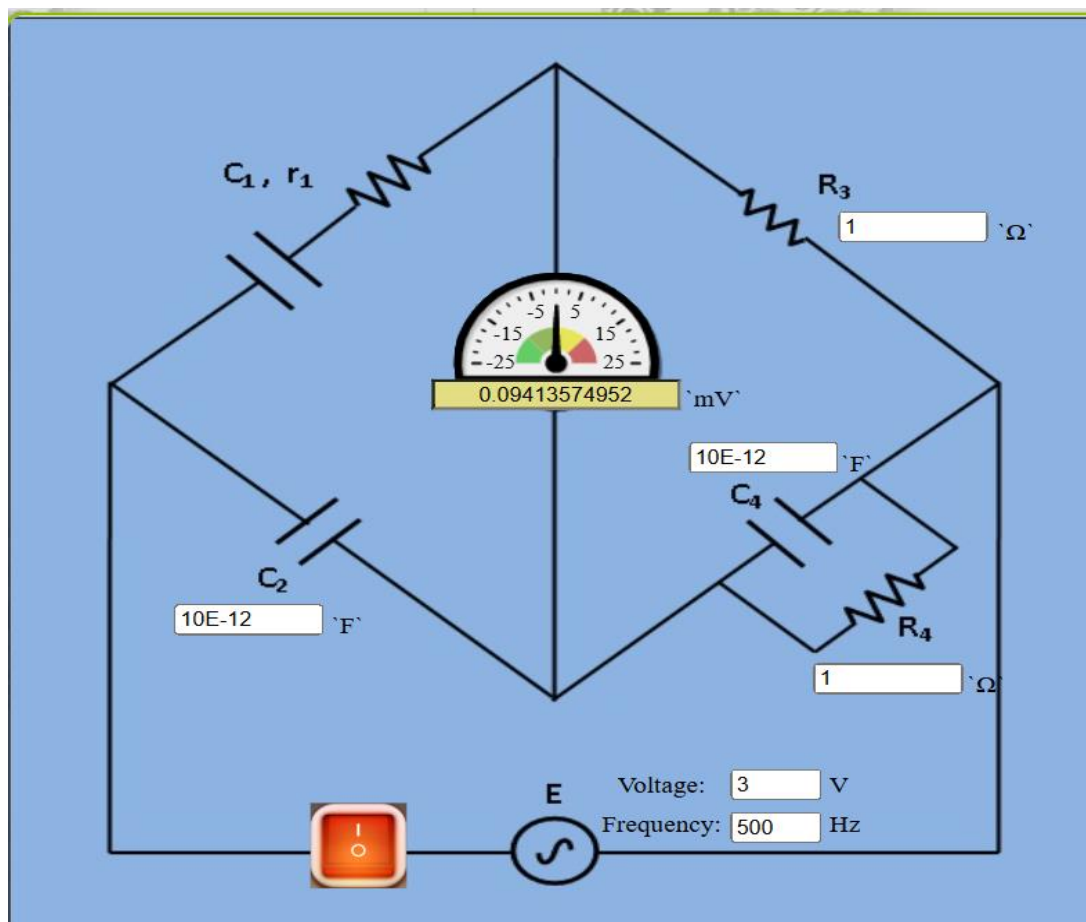


Fig. 1. Circuit diagram of experimental set-up for Capacitance measurement by Schering Bridge.

- 1) Apply Supply voltage from the signal generator with arbitrary frequency. ($V = 3\text{V}$). Also set the unknown Capacitance value from 'Set Capacitor Value' tab.
- 2) Then switch on the supply to get millivoltmeter deflection.
- 3) Choose the values of C_2 , C_4 , R_3 and R_4 from the capacitance and resistance box. Vary the values to some particular values to achieve "NULL".
- 4) Observe the millivoltmeter pointer to achieve "NULL".
- 5) If "NULL" is achieved, switch to 'Measure Capacitor Value' tab and click on 'Simulate'. Observe the calculated values of unknown capacitance (C_1) and its internal resistance (r_1).
- 6) Also observe the Dissipation factor of the unknown capacitor which is defined as

$$\omega * C * r \text{ Where, } \omega = 2\pi f$$

SIMULATION



The current voltmeter reading is: 0.094135749E mv. Now simulate to get:

Capacitor value (in uF): 0.000010000

Resistance value (in Ω): 1.0000

Dissipation Factor: 3.1400e-8

Simulate

RESULT:

Thus, the unknown capacitance is found using Schering bridge.